Kaushikee Karmakar

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Education

Asansol Engineering College (AEC) Aug 2024 – Mar 2028
Bachelor of Computer Applications (BCA) *Year 1 CGPA: 9.48* *Year 2 CGPA: 9.09*
SGPA: **1st Sem** 9.41 — **2nd Sem** 9.55 — **3rd Sem** 9.29 — **4th Sem** 8.89

Projects

Backyard Sweep: Mosquito Breeding-Site Detector Oct 2026
YOLOv8, Ultralytics, Render [GitHub] [Write-up]
- Trained YOLOv8n on VisText-Mosquito to detect tires, vases, bottles, coconut shells and drain inlets, reaching 0.926 mAP50 on the test split; deployed as a public web app (Hacktoberfest 2026).
- Field-tested on my own photos and documented where the model fails (misclassified objects, missed drains) instead of only reporting the headline metric.

PRISM-TC: Tropical Cyclone Intensity Classification SIH 2026
PyTorch, timm, EfficientNet-B0 [GitHub]
- Evaluation Lead on a 5-member Smart India Hackathon team (SIH260070) classifying cyclone intensity (5 imbalanced classes) from satellite images (TCIR + IBTrACS).
- Ran the held-out test evaluation (372 images): accuracy, macro-F1, per-class metrics, confusion matrix and calibration; recommended the original model (60.2% acc., 0.52 macro-F1) over the upgrade (55.9%, 0.41) after it regressed on the highest-risk failure mode.

TaskFlow: Task & Focus Workspace 2026
Flask, SQLite, JavaScript, Gunicorn [GitHub]
- Built a responsive app with task CRUD, categories, priorities, due dates, estimates, filters, a dashboard of active, due-today and overdue counts, and a 25/15/5-minute focus timer.
- Added a transparent rules-based planner that orders tasks by deadline, priority and effort, plus a health-check endpoint for deployment.

Stuck Doctor Oct 2026
Gemma, Ollama, Gradio [GitHub] [Write-up]
- Built a local-LLM app that helps people who keep putting off the doctor actually book an appointment (Hacktoberfest Weekend Challenge).

Mining Gas Alarm System Mar 2026
Hardware model, AEC Hardware Club (IHMMC-2026)
- Designed and exhibited a hardware model at the college-level Innovative Hardware Model Making Competition.

In progress: Student Dropout web app — healthcare-focused AI/ML project

Training

SparkIIT AI Training & Internship Program Jul 2026 batch, 3 months
Participant, AI Training (powered by Wipro DICE ID) *[GitHub]*
- Capstone project: built an end-to-end NLP document classifier on the 20 Newsgroups corpus (6 topics, 5,668 documents): text cleaning, TF-IDF with unigrams and bigrams, no train/test leakage.
- Compared Naive Bayes, Logistic Regression, Linear SVM and a MiniLM embedding baseline; tuned the best with GridSearchCV to reach **91.1% accuracy** (0.91 weighted F1) on 1,131 held-out documents.
- Deployed as a Streamlit app on Render with confidence scores and the words driving each prediction; wrote a 3-page report with error analysis and limitations.

Technical Skills

Programming Languages: Python, Java, C, SQL, HTML, CSS, JavaScript
ML & AI: PyTorch, scikit-learn, Ultralytics YOLOv8, timm, NLP (TF-IDF, sentence-transformers), model evaluation, Generative AI & prompting, Ollama
Frameworks & Tools: Flask, Streamlit, Gradio, SQLite, Git, GitHub, Render, Google Colab, Jupyter, Power BI
Other Skills: Data Structures & Algorithms (Python, Java), data analysis, public speaking, creative writing, sketching

Certifications

NASSCOM: Gen AI Tools (Aug 2026) — **SparkIIT**: Growth and Operation (Jun 2026) — **Intel AI for All**: AI Aware & AI Appreciate (Oct 2025)
Forage Job Simulations: Tata GenAI Data Analytics, Deloitte Data Analytics, Data Labeling (Jan 2026)
HP LIFE: Critical Thinking in the AI Era, AI for Business Professionals (Jan 2026) — **Simplilearn**: SQL Analytics & BI on Databricks (Jan 2026) — **Udemy**: Mastering Power BI (Nov 2025)

Activities & Achievements

- **Hacktoberfest 2026 Completionist:** published 2 project write-ups on DEV Community.
- **GDG on Campus AEC:** completed the 3-Day Web Bootcamp (Nov 2025), building a live game in HTML, CSS and JavaScript.
- **NDLI Club National Essay Writing Competition (IIT Kharagpur):** ranked 6th at my college (Aug 2025).
- **NSS volunteer, AEC:** Swachhata Hi Seva 2025 cleanliness drive; Lit Fest 2025 participant.